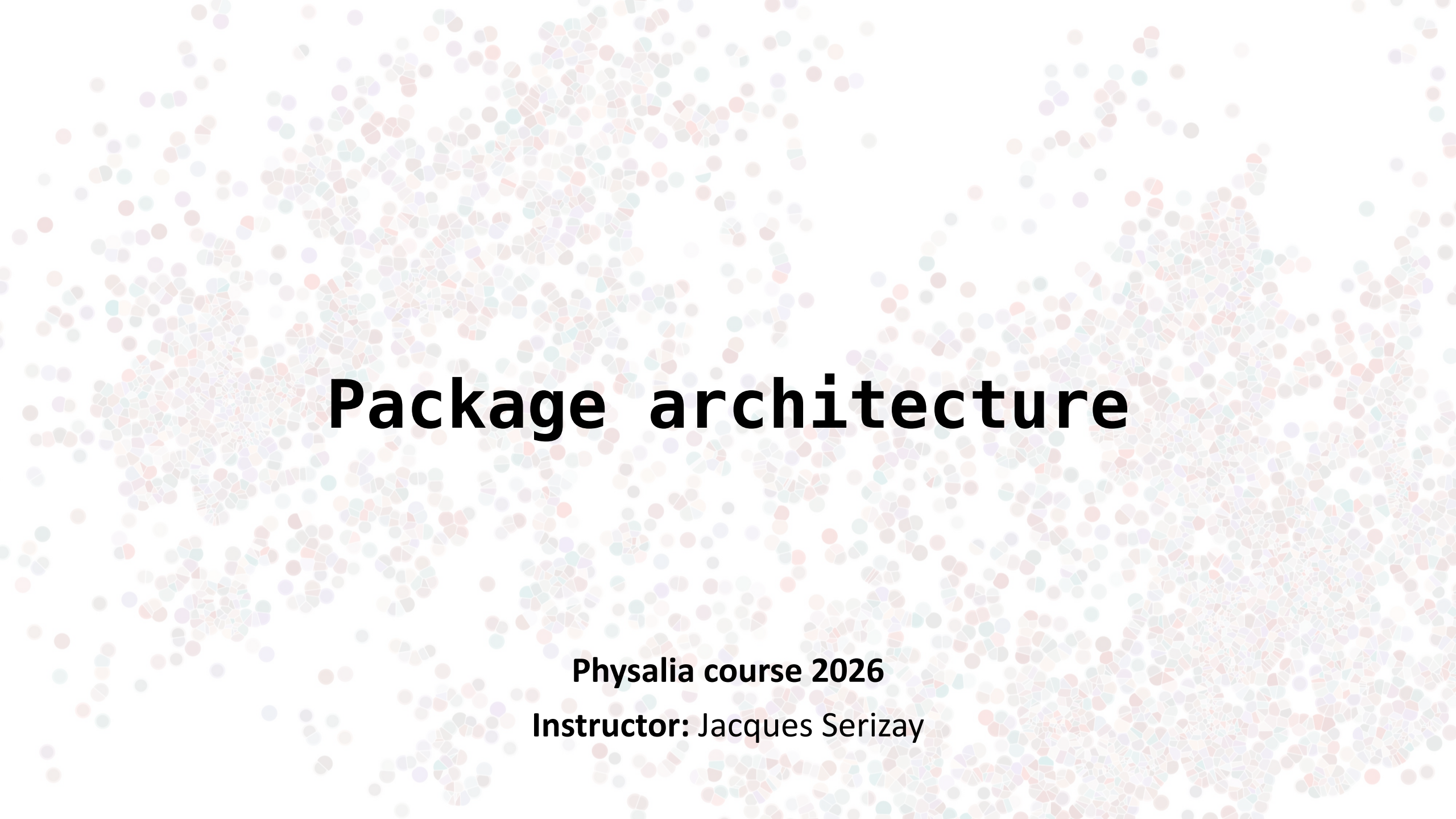




Package architecture

Physalia course 2026

Instructor: Jacques Serizay

What are packages

- R “core” is already a mix of several packages !
 - base
 - compiler
 - datasets
 - graphics
 - grDevices
 - grid
 - methods
 - parallel
 - splines
 - stats
 - stats4
 - tcltk
 - tools
 - translations
 - utils

What are packages

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 - compiler
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 - grid
 - methods
 - parallel
 - splines
 - stats
 - stats4
 - tcltk
 - tools
 - translations
 - utils

R preferred extension mechanism: new functionalities (through new functions, but not only!)

What are packages

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- base
- compiler
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- grid
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- parallel
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- stats
- stats4
- tcltk
- tools
- translations
- utils

22922

CRAN PACKAGES

<https://rdrr.io/>

2130

BIOCONDUCTOR PACKAGES

2206

R-FORGE PACKAGES

85749

GITHUB PACKAGES

R preferred extension mechanism: new functionalities (through new functions, but not only!)

Standard package source: the bare minimum

```
myPackage/
```

```
R/
```

```
  functions.R
```

```
  utils.R
```

```
DESCRIPTION
```

```
NAMESPACE
```

Standard package source: more complete structure

myPackage/

data/

man/

inst/

tests/

vignettes/

R/

functions.R

utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENSE

DESCRIPTION

myPackage/
data/
man/
inst/
tests/
vignettes/
R/
functions.R
utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENSE

Package: HiContacts
Title: HiContacts: R interface to cool files
Version: 0.99.11
Date: 2022-08-16
Authors@R:
 c(person(given = "Jacques",
 family = "Serizay",
 role = c("aut", "cre"),
 email = "jacquesserizay@gmail.com",
 comment = c(ORCID = "0000-0002-4295-0624")))
Description: HiContacts: R interface to (m)cool files and other Hi-C
 processed file formats. HiContacts provides a collection of tools to
 analyse and visualize Hi-C datasets. It can import data from pairs
 or (m)cool files.
License: MIT + file LICENSE
URL: <https://github.com/js2264/HiContacts>
BugReports: <https://github.com/js2264/HiContacts/issues>
Depends:
 R (>= 4.2)
Imports:
 HiContactsData
Suggests:
 rmarkdown
biocViews:
 HiC
Encoding: UTF-8
VignetteBuilder: knitr
LazyData: false
Roxygen: list(markdown = TRUE)
RoxygenNote: 7.2.1

myPackage/
data/
man/
inst/
tests/
vignettes/
R/
functions.R
utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENSE

Package ‘HiContacts’

October 14, 2022

Title HiContacts: R interface to cool files
Version 0.99.11
Date 2022-08-16
Description HiContacts: R interface to (m)cool files and other Hi-C processed file formats. HiContacts provides a collection of tools to analyse and visualize Hi-C datasets. It can import data from pairs or (m)cool files.
License MIT + file LICENSE
URL <https://github.com/js2264/HiContacts>
BugReports <https://github.com/js2264/HiContacts/issues>
Depends R (>= 4.2)
Imports HiContactsData, InteractionSet, GenomicInteractions, GenomicRanges, IRanges, GenomeInfoDb, S4Vectors, BiocGenerics, methods, rhdf5, Matrix, vroom, tibble, tidyr, dplyr, glue, stringr, reticulate, ggplot2, ggrastr, scales
Suggests cowplot, testthat (>= 3.0.0), BiocStyle, knitr, rmarkdown
biocViews HiC, DNA3DStructure, DataImport
Encoding UTF-8
VignetteBuilder knitr
LazyData false
Roxygen list(markdown = TRUE)
RoxygenNote 7.2.1
git_url <https://git.bioconductor.org/packages/HiContacts>
git_branch master
git_last_commit 6ccb53e
git_last_commit_date 2022-10-13
Date/Publication 2022-10-14
Author Jacques Serizay [aut, cre] (<<https://orcid.org/0000-0002-4295-0624>>)
Maintainer Jacques Serizay <jacquesserizay@gmail.com>

NAMESPACE

myPackage/

data/

man/

inst/

tests/

vignettes/

R/

functions.R

utils.R

DESCRIPTION

NAMESPACE

README.md

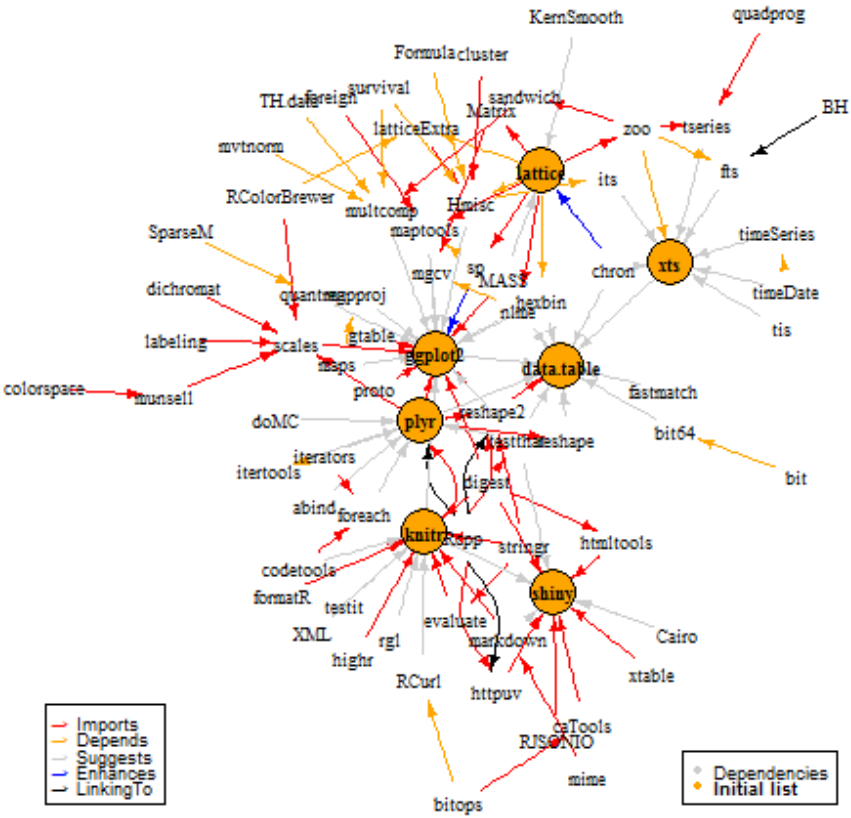
NEWS

LICENSE

Generated by roxygen2: do not edit by hand

```
export(APA)
export(cool2gi)
export(cool2seqinfo)
export(detrend)
export(fetchCool)
export(formatCoords)
export(getAnchors)
export(getCounts)
export(getHicStats)
export(getPs)
export(ggMatrix)
export(ggPs)
export(ggtheme_HiContacts)
export(ggtheme_HiContacts_tracks)
export(gi2cm)
export(lsCoolFiles)
export(lsCoolResolutions)
export(pairs2gi)
export(peekCool)
export(plotMatrix)
export(splitCoords)
import(BiocParallel)
import(InteractionSet)
import(dplyr)
import(ggplot2)
import(ggrastr)
import(rhdf5)
import(scales)
import(stringr)
import(tibble)
import(tidyr)
import(tools)
import(zeallot)
importFrom(GenomeInfoDb,Seqinfo)
importFrom(GenomeInfoDb,seqlengths)
importFrom(GenomicRanges,GRanges)
importFrom(GenomicRanges,end)
importFrom(GenomicRanges,findOverlaps)
importFrom(GenomicRanges,mcols)
importFrom(GenomicRanges,resize)
importFrom(GenomicRanges,seqnames)
importFrom(GenomicRanges,start)
```

Package dependency graph



myPackage/
data/
man/
inst/
tests/
vignettes/
R/
functions.R
utils.R
DESCRIPTION
NAMESPACE
README.md
LICENSE

```
# HiContacts

HiContacts provides tools to import `(m)cool` matrices in R and work with them there.

It creates a new `Contacts` class of objects, built on pre-existing Bioconductor objects, namely `InteractionSet`, `GenomicInteractions` and `ContactMatrix` (Lun, Perry & Ing-Simmons, F1000Research 2016), and provides **analytical** and **visualization** tools to investigate contact maps.

## Installation

`HiContacts` will soon be available in Bioconductor. To install the current release use:

```r
if (!requireNamespace("BiocManager", quietly = TRUE))
 install.packages("BiocManager")
BiocManager::install("HiContacts")
```

For now, `HiContacts` can be installed from Github as follows:

```r
install.packages("devtools")
devtools::install_github("js2264/HiContacts")
library(HiContacts)
```

## Citation

If you are using `HiContacts` in your research, please cite:

> Serizay J (2022). _HiContacts: HiContacts: R interface to cool files_.  

> R package version 0.99.0  

> <https://github.com/js2264/HiContacts>.

## How to use `HiContacts`

`HiContacts` includes a introduction vignette where its usage is illustrated. To access the vignette, please use:

```r
vignette('HiContacts')
```

## Overview

### Import a .(m)cool file as `Contacts`
```

HiContacts

lifecycle experimental R-CMD-check-bioc passing pkgdown passing

HiContacts provides tools to import `(m)cool` matrices in R and work with them there.

It creates a new `Contacts` class of objects, built on pre-existing Bioconductor objects, namely `InteractionSet`, `GenomicInteractions` and `ContactMatrix` (Lun, Perry & Ing-Simmons, F1000Research 2016), and provides **analytical** and **visualization** tools to investigate contact maps.

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How to use HiContacts

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```
vignette('HiContacts')
```

Overview

Import a `.(m)cool` file as `Contacts`

- myPackage/
- data/
- man/
- inst/
- tests/
- vignettes/
- R/
- functions.R
- utils.R
- DESCRIPTION
- NAMESPACE
- README.md
- NEWS
- LICENSE

```
# periodicDNA 0.3.2 (dev)

* IMPORTANT:
- Implemented data-raw for reproducibility

* MINOR:
- Changed xlim of norm. distr. plot in plotPeriodicityResults()

# periodicDNA 0.3.1 (2020-05-05)

* IMPORTANT:
- rollmean(k=3) is now applied *before* normalisation *as well*,
  on the raw distribution vector
- plotPeriodicityResults() output returns one single plot (with cowplot)
- getPeriodicityTrack() now returns the Rle
- Improved plotting functions -now show shuffled for plotPeriodicityResults()
- Added ggplot2 theming

* MINOR:
* Changed many variable names (all to snake_case)
* sampleGRanges is now full-fledged function
  (GRanges, DNASTringSet, character and BSgenome methods)
* sampleGenome is an alias for sampleGRanges.character
* Added sacCer3 to getPeriodicity BSgenomes
* Added DNASTring method for getPeriodicity
* Added a vignette describing the internal steps
* Clarified user-level functions in README
* Added ce11_TSSs data
* Renamed generateperiodicitytrack as getPeriodicityTrack
* Renamed variables in getFPI and getPeriodicity
* Created a utility char2BSgenome()

# periodicDNA 0.3.0 (2020-05-03)

* Added tests
* Added getFPI function
* cleaned-up functions
* cleaned-up function dependencies
* Added toy data
* Added vignette

# periodicDNA 0.2.1 (2020-03-04)

* Added Travis build check
* Simplified README.md
```

- myPackage/
- data/
- man/
- inst/
- tests/
- vignettes/
- R/
- functions.R
- utils.R
- DESCRIPTION
- NAMESPACE
- README.md
- NEWS
- LICENSE

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=====

Version 3, 29 June 2007
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Some devices are designed to deny users access to install or run modified versions of the software inside them. although the manufacturer can do so. This is fundamentally

Function files

myPackage/

data/

man/

inst/

tests/

vignettes/

R/

functions.R

utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENSE

```
#' Parsing (m)cool files
#'
#' These functions are the workhorse internal functions used to import
#' a `.(m)cool` file as GenomicInteractions (wrapped into a `Contacts` object
#' by `Contacts()` function).
#'
#' @param file file
#' @param resolution resolution
#' @param balanced import balancing scores
#' @return anchors from (m)cool, stored as a GRanges
#'
#' @importFrom GenomicRanges GRanges
#' @importFrom GenomicRanges seqnames
#' @importFrom GenomicRanges start
#' @importFrom GenomicRanges end
#' @importFrom IRanges IRanges
#' @rdname parse

getAnchors <- function(file, resolution = NULL, balanced = "cooler") {
  bins <- fetchCool(file, "bins", resolution)
  anchors <- GenomicRanges::GRanges(
    bins$chr,
    IRanges::IRanges(bins$start + 1, bins$end),
    bin_id = seq_along(bins$chr),
    seqinfo = cool2seqinfo(file, resolution)
  )
  names(anchors) <- paste(GenomicRanges::seqnames(anchors), GenomicRanges::start(anchors), GenomicRanges::end(anchors), sep
= " ")
  if (~"weight" %in% names(bins) & {
    balanced == "cooler" | balanced == TRUE
  }) {
    anchors$weight <- as.numeric(bins$weight)
  } else {
    weight <- 1
  }
  return(anchors)
}
```

myPackage/

data/

man/

functions.Rd

inst/

tests/

vignettes/

R/

functions.R

utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENCE

```
% Generated by roxygen2: do not edit by hand
% Please edit documentation in R/parse.R
\name{getAnchors}
\alias{getAnchors}
\alias{getCountsFromPair}
\alias{getCounts}
\alias{fetchCool}
\alias{lsCoolFiles}
\alias{lsCoolResolutions}
\alias{peekCool}
\alias{cool2seqinfo}
\alias{cool2gi}
\alias{gi2cm}
\alias{cm2matrix}
\alias{pairs2gi}
\title{Parsing (m)cool files}
\usage{
  getAnchors(file, resolution = NULL, balanced = "cooler")
}
\arguments{
  \item{file}{pairs file: \verb{<readname>\\t<chr1>\\t<start1>\\t<chr2>\\t<start2>}}

  \item{resolution}{resolution}

  \item{balanced}{import balancing scores}
}
\value{
  anchors from (m)cool, stored as a GRanges
}
\description{
  These functions are the workhorse internal functions used to import
  a \verb{.(m)cool} file as GenomicInteractions (wrapped into a \code{Contacts} object
  by \code{Contacts()} function).
}
```

myPackage/

data/

man/

functions.Rd

inst/

tests/

vignettes/

R/

functions.R

utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENCE

pdf documentation once package is published

getAnchors

Parsing (m)cool files

Description

These functions are the workhorse internal functions used to import a .(m)cool file as GenomicInteractions (wrapped into a Contacts object by Contacts() function).

Usage

getAnchors(file, resolution = NULL, balanced = "cooler")

Arguments

| | |
|------------|--|
| file | pairs file: <readname>\t<chr1>\t<start1>\t<chr2>\t<start2> |
| resolution | resolution |
| balanced | import balancing scores |

Value

anchors from (m)cool, stored as a GRanges

myPackage/

data/

man/

functions.Rd

inst/

tests/

vignettes/

R/

functions.R

utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENCE

?getAnchors

getAnchors

package:HiContacts

R Documentation

[Parsing \(m\)cool files](#)

[Draw](#)

[Design](#)

[Transitions](#)

[Animations](#)

[Slide](#)

[Description:](#)

These functions are the workhorse internal functions used to import a .(m)cool file as GenomicInteractions (wrapped into a 'Contacts' object by 'Contacts()'

[Usage:](#)

```
getAnchors(file, resolution = NULL, balanced = "cooler")
```

[Arguments:](#)

file: pairs file: <readname>\t<chr1>\t<start1>\t<chr2>\t<start2>

resolution: resolution

balanced: import balancing scores

[Value:](#)

anchors from (m)cool, stored as a GRanges

Source/Bundle/Binary/Installed/Loaded: késako?

myPackage/

data/

man/

inst/

tests/

vignettes/

R/

functions.R

utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENSE

Source/Bundle/ Binary /Installed/Loaded: késako?

myPackage/

data/

man/

inst/

tests/

vignettes/

R/

functions.R

utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENSE

Source

Source/Bundle/ Binary /Installed/Loaded: késako?

```
myPackage/
```

data/

man/

inst/

tests/

vignettes/

R/

functions.R

utils.R

DESCRIPTION

NAMESPACE

README.md

NEWS

LICENSE

r-lib / devtoolsPublic

<> CodeIssues12Pull requests1ActionsProjectsWikiSecurityInsights

main40 branches48 tagsGo to fileAdd fileCode

jennybcIncrement version number to 2.4.5.9000aa3f88b23 days ago3,463 commits

| | | |
|------------------|--|---------------|
| .github | Add linting action (#2431) | 5 months ago |
| R | Set package in test_active_file() (#2471) | last month |
| inst | spelling::update_wordlist() | 4 months ago |
| man-roxygen | Clarify documentation of the pkg argument | 3 years ago |
| man | Surface unused imports with check(cran = TRUE) (#2461) | 3 months ago |
| pkgdown/favicon | Minor improvements to website (#2256) | 2 years ago |
| revdep | Check revdeps | 26 days ago |
| tests | Set package in test_active_file() (#2471) | last month |
| vignettes | Remove last vestiges of the master branch | 12 months ago |
| .Rbuildignore | Rbuildignore submission file | 4 months ago |
| .gitattributes | enable union merge for NEWS.md file | 9 years ago |
| .gitignore | Ignore some stuff | 4 months ago |
| .lintr | Add linting action (#2431) | 5 months ago |
| DESCRIPTION | Increment version number to 2.4.5.9000 | 23 days ago |
| LICENSE | Re-license as MIT (#2334) | 2 years ago |
| LICENSE.md | Re-license as MIT (#2334) | 2 years ago |
| MAINTENANCE.md | Add maintenance document | 12 months ago |
| NAMESPACE | Quiet check (#2447) | 4 months ago |
| NEWS.md | Increment version number to 2.4.5.9000 | 23 days ago |
| README.md | Update cheatsheet links | 4 months ago |
| _pkgdown.yml | Switch to pkgdown "auto" development mode | 11 months ago |
| codecov.yml | Convert to using GitHub Actions | 3 years ago |
| cran-comments.md | Check revdeps | 26 days ago |
| devtools.Rproj | RStudio update | 5 months ago |

AboutTools to make an R developer's life easier

devtools.r-lib.orgpackage-creation

ReadmeUnknown, MIT licenses foundCode of conduct2.2k stars127 watching750 forks

Releases35

devtools 2.4.5Latest23 days ago

+ 34 releases

Packages

No packages published

Contributors145

+ 134 contributors

Environments1

github-pagesActive

Languages

Source/Bundle/ Binary /Installed/Loaded: késako?

myPackage/

data/

man/

inst/

tests/

vignettes/

R/

functions.R

utils.R

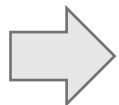
DESCRIPTION

NAMESPACE

README.md

NEWS

LICENSE



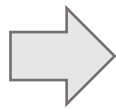
myPackage.tar.gz

Bundle

Source/Bundle/ Binary /Installed/Loaded: késako?

myPackage/
data/
man/
inst/
tests/
vignettes/
R/

functions.R
utils.R
DESCRIPTION
NAMESPACE
README.md
NEWS
LICENSE



myPackage.tar.

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<> CodeIssues12Pull requests1ActionsProjectsWikiSecurityInsights

main40 branches48 tagsGo to fileAdd fileCode

jennybcIncrement version number to 2.4.5.9000aa3f88b23 days ago3,463 commits

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| devtools.Rproj | RStudio update | 5 months ago |

About

Tools to make an R developer's life easier

[devtools.r-lib.org](#)

rpackage-creation

Readme

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Code of conduct

2.2k stars

127 watching

750 forks

Releases35

devtools 2.4.5Latest23 days ago

+ 34 releases

Packages

No packages published

Contributors145

+ 134 contributors

Environments1

github-pagesActive

Languages

Source/Bundle/ Binary /Installed/Loaded: késako?

myPackage/
data/
man/
inst/
tests/
vignettes/
R/

functions.R
utils.R
DESCRIPTION
NAMESPACE
README.md
NEWS
LICENSE



23 days ago
jennybc
v2.4.5
4a7ba32
Compare

devtools 2.4.5 Latest

- check(cleanup =) was deprecated in devtools v1.11.0 (2016-04-12) and was made defunct in v2.4.4 (2022-07-20). The documentation is more clear now about recommended alternatives.
- check(check_dir = NULL) is the new default, to align with the default behaviour of the underlying rcmdcheck::rcmdcheck().
- check(cran = TRUE) sets the env var _R_CHECK_PACKAGES_USED_IGNORE_UNUSED_IMPORTS_ to FALSE, in order to surface the "Namespace in Imports field not imported from" NOTE. This only applies to R >= 4.2, due to favorable changes in the behaviour of R CMD check --as-cran (#2459).
- test_active_file() passes the package onto to testthat so it can correctly set the TESTTHAT_PKG envvar (#2470).
- test() and test_active_file() once again work with testthat itself.

▼ Assets 2

Source code (zip) 23 days ago

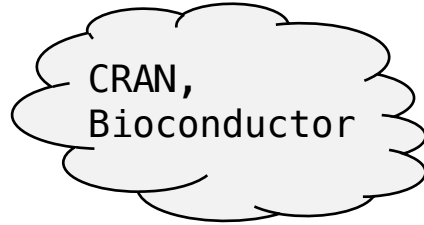
Source code (tar.gz) 23 days ago

😊

Source/Bundle/ Binary /Installed/Loaded: késako?

myPackage/
data/
man/
inst/
tests/
vignettes/
R/
functions.R
utils.R
DESCRIPTION
NAMESPACE
README.md
NEWS
LICENSE

➡ myPackage.tar.gz



Binary

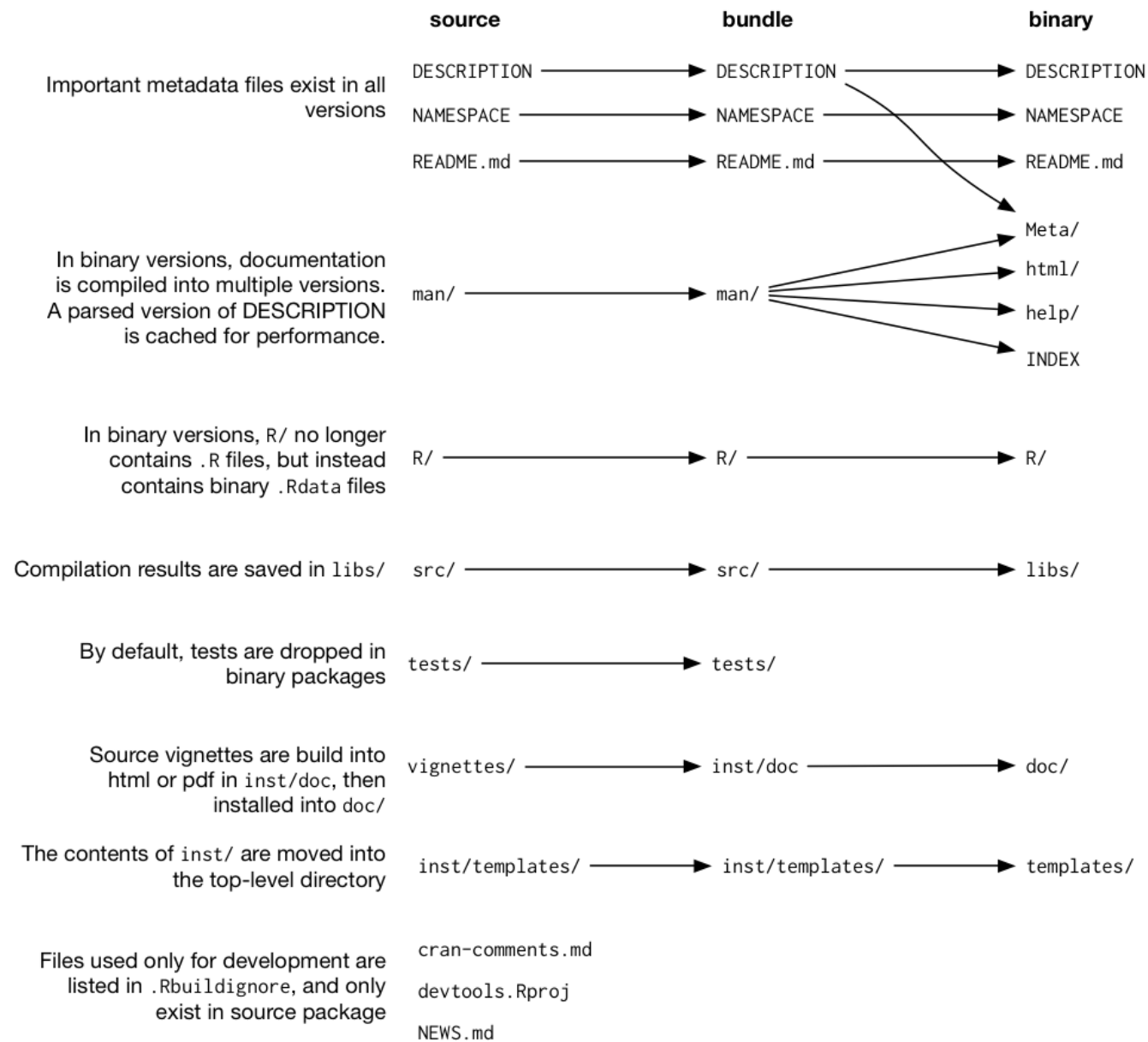
Package Archives

Follow [Installation](#) instructions to use this package in your R session.

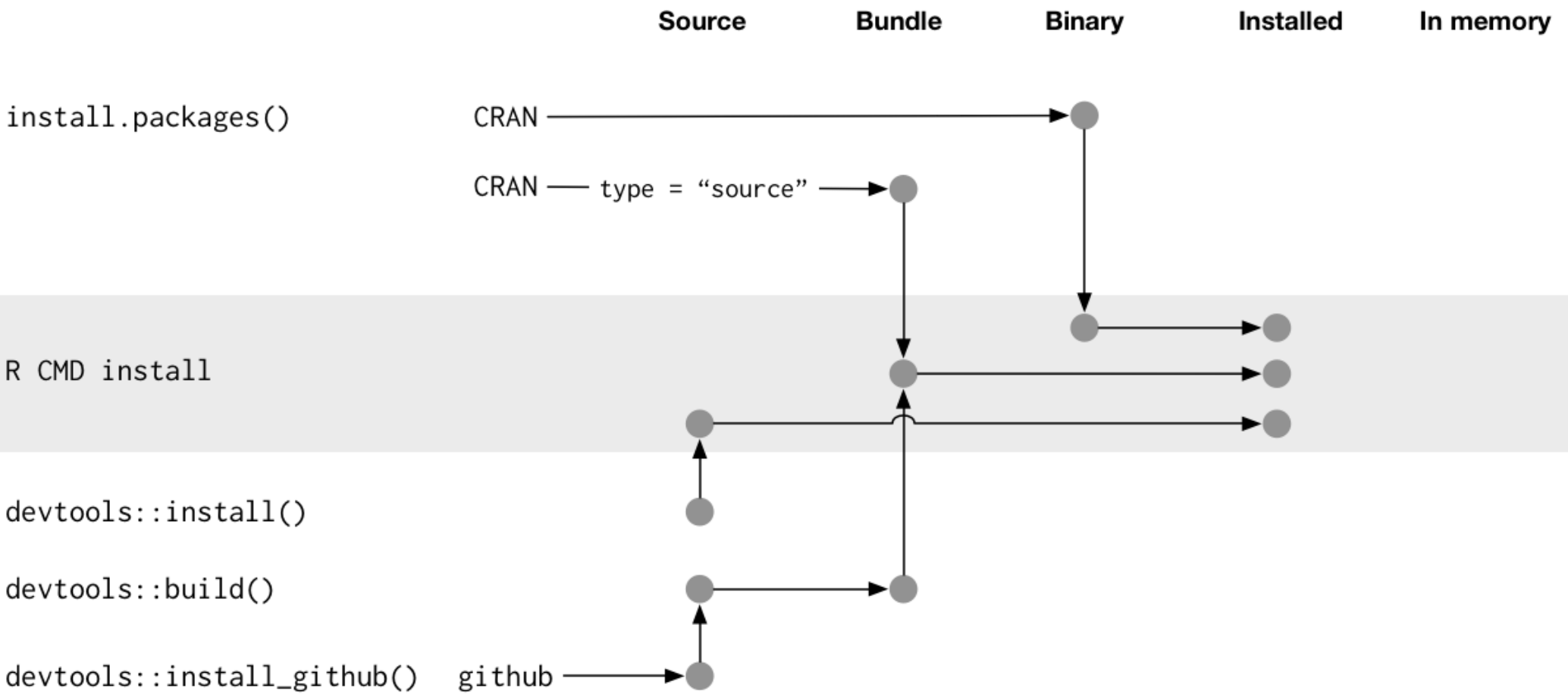
| | |
|-----------------------|---|
| Source Package | S4Vectors 0.34.0.tar.gz |
| Windows Binary | S4Vectors 0.34.0.zip |
| macOS Binary (x86_64) | S4Vectors 0.34.0.tgz |



Source/Bundle/ Binary /Installed/Loaded: késako?



Source/Bundle/ Binary /Installed/Loaded: késako?



Read more...

<https://r-pkgs.org/Structure.html>